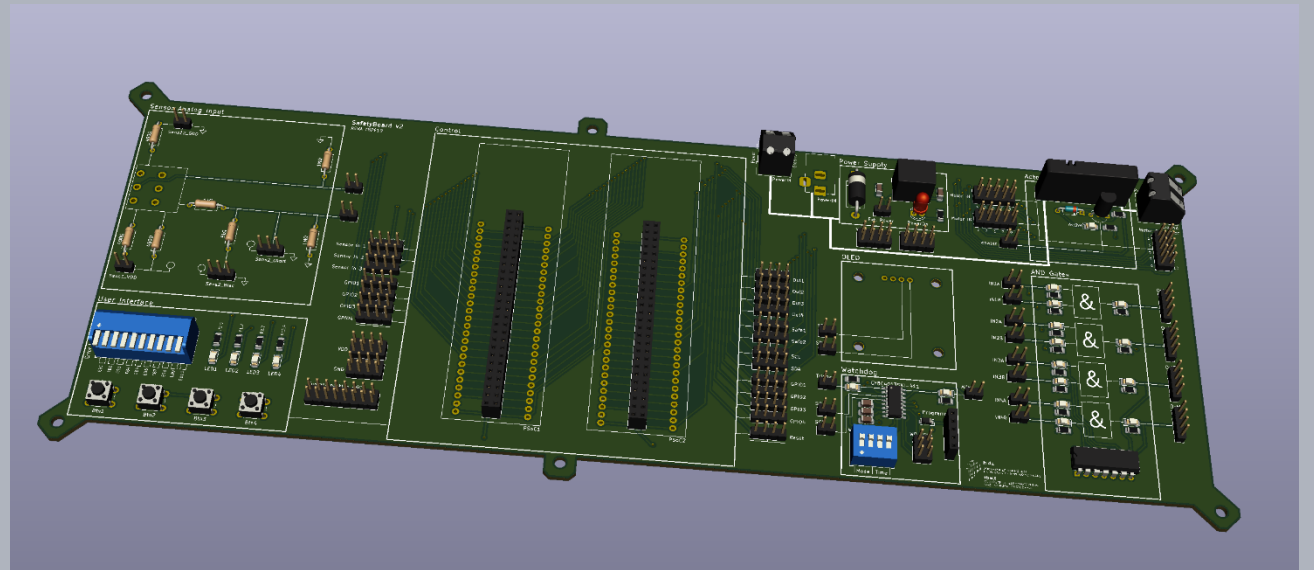


Safety in Embedded Control Systems

Introduction

Prof. Dr.-Ing. Peter Fromm



Content

- Sometimes, things go terribly wrong...
- What means functional safety?
- Risks and Hazards
- Safety in Embedded Control Systems
- Typical problems
- Safety = Architecture + Process
- Safety versus (?) Security and other trends



Source: Wikipedia

Therac 25 (1985 – 1987)

- The Therac-25 was a computer-controlled radiation therapy machine produced by Atomic Energy of Canada Limited (AECL) in 1982
- It caused 6 dangerous accidents when the high current beam was delivered directly to patients
- Software Bug One allowed the operator to incorrectly select X-ray mode before quickly changing to electron mode, which allowed the electron beam to be set for X-ray mode without the X-ray target being in place.
- A second fault allowed the electron beam to activate during field-light mode, during which no beam scanner was active or target was in place.
- Previous models had hardware interlocks to prevent such faults, but the Therac-25 had removed them, depending instead on software checks for safety.



Source: Wikipedia

Root cause analysis (engineering)

The researchers also found several engineering issues:

- The failure occurred only when a particular nonstandard sequence of keystrokes was entered on the VT-100 terminal which controlled the PDP-11 computer: an "X" to (erroneously) select 25 MeV photon mode followed by "cursor up", "E" to (correctly) select 25 MeV Electron mode, then "Enter", all within eight seconds.
- The design did not have any hardware interlocks to prevent the electron-beam from operating in its high-energy mode without the target in place.
- The engineer had reused software from older models. Such methods manifest in so called cargo cult programming where there is blind reliance on previously created code that is poorly understood and may or may not be applicable. These models had hardware interlocks that masked their software defects. Those hardware safeties had no way of reporting that they had been triggered, so there was no indication of the existence of faulty software commands.
- The hardware provided no way for the software to verify that sensors were working correctly (see open-loop controller). The table-position system was the first implicated in Therac-25's failures; the manufacturer revised it with redundant switches to cross-check their operation.
- The equipment control task did not properly synchronize with the operator interface task, so that race conditions occurred if the operator changed the setup too quickly. This was missed during testing, since it took some practice before operators were able to work quickly enough to trigger this failure mode.
- The software set a flag variable by incrementing it, rather than by setting it to a fixed non-zero value. Occasionally an arithmetic overflow occurred, causing the flag to return to zero and the software to bypass safety checks.

Source: Wikipedia

Root cause analysis (process)

Researchers who investigated the accidents found several contributing causes. These included the following institutional causes:

- AECL did not have the software code independently reviewed.
- AECL did not consider the design of the software during its assessment of how the machine might produce the desired results and what failure modes existed. These form parts of the general techniques known as reliability modeling and risk management.
- The system noticed that something was wrong and halted the X-ray beam, but merely displayed the word "MALFUNCTION" followed by a number from 1 to 64. The user manual did not explain or even address the error codes, so the operator pressed the P key to override the warning and proceed anyway.
- AECL personnel, as well as machine operators, initially did not believe complaints. This was likely due to overconfidence.
- AECL had never tested the Therac-25 with the combination of software and hardware until it was assembled at the hospital.

Source: Wikipedia

Lufthansa Flight 2904 landing in Warszawa (1993)

- After landing, the plane could not brake and crashed behind the landing zone, catching fire.
- 2 passengers got killed, 54 wounded.

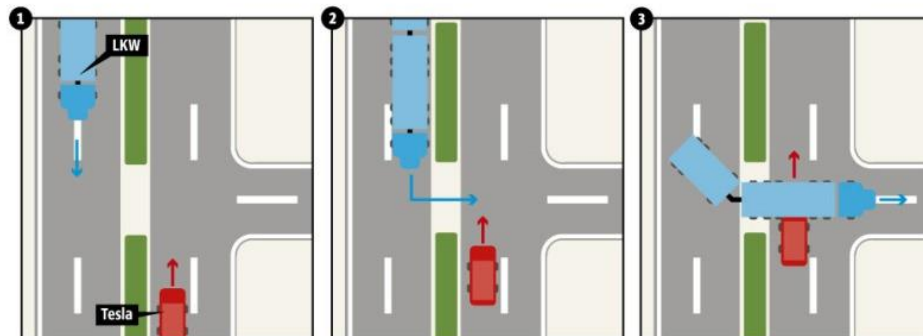
Problem

- The smart flight-by-wire system only allowed reverse throttle after the plane touchdown.
- Touchdown was detected by 12t weight on the wheels AND wheels turning in driving direction.
- Due to aqua planning, the wheels did not turn.



Tesla truck crash (2016)

- In 2016, a Tesla S crashed into a truck, although the very advanced anti collision system was turned on.
- The sensors system was realised in a redundant manner: cameras, radar and ultrasonic sensors were in place.
- Camera: The truck was hit in front of a light sky
- Ultrasonic: only short distance and not reliable
- Radar: Might have misjudged the truck with a traffic sign bridge.



Lion Air Crash on 29.11.2018 and Ethiopian-Airlines Crash March 2019



Boeing 737 MAX

Chain of wrong (criminal) descisions...

- Starting point: Fuel consumption reduction
- This leads to larger engines, which cannot be mounted at the optimal position
- Increased risk of uncontrolled climbing
- Introduction of the Safety Feature MCAS



<https://spectrum.ieee.org/aerospace/aviation/how-the-boeing-737-max-disaster-looks-to-a-software-developer>

How the new Max flight-control system (MCAS) operates to prevent a stall

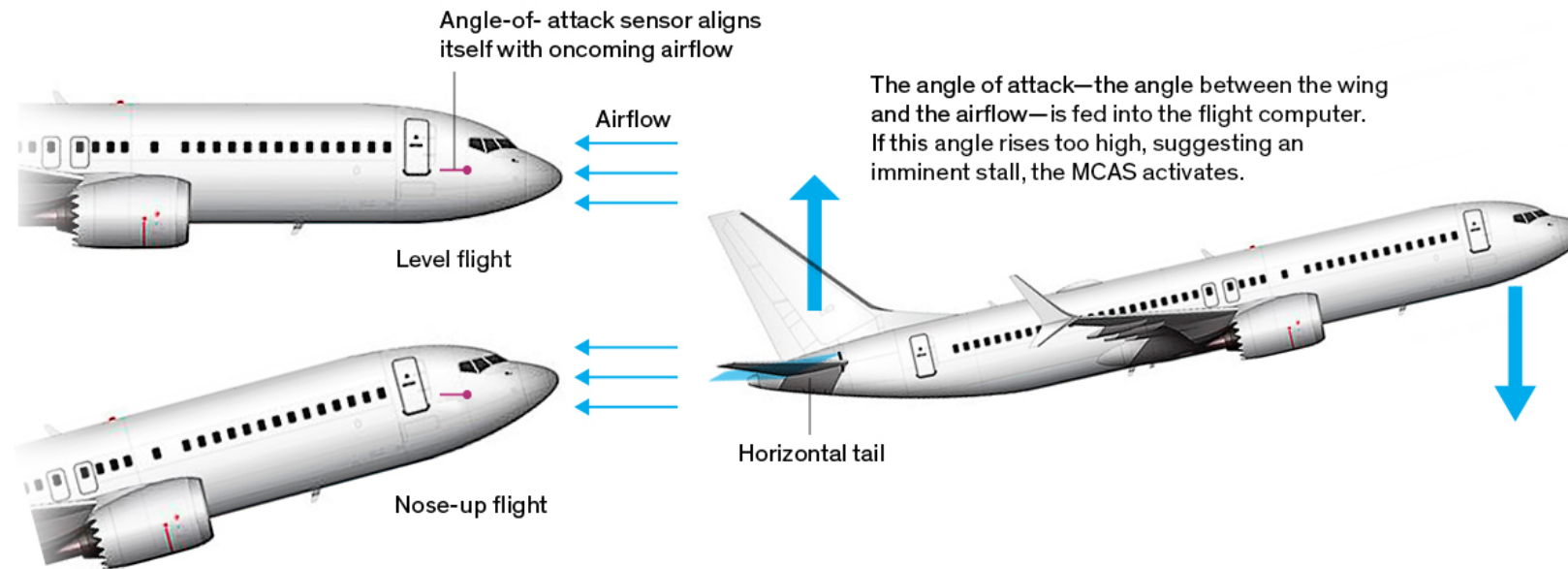


Illustration: Norebbo.com

The antistall system depended crucially on sensors that are installed on each side of the airliner—but the system consulted only the sensor on one side.

<https://spectrum.ieee.org/aerospace/aviation/how-the-boeing-737-max-disaster-looks-to-a-software-developer>

Chain of Problems

Wrong angle
declared in
specification

QA by Boeing
instead of FAA

So Boeing produced a dynamically unstable airframe, the 737 Max. That is big strike No. 1. Boeing then tried to mask the 737's dynamic instability with a software system. Big strike No. 2. Finally, the software relied on systems known for their unreliability. Big strike No. 3. **Failure detection only optional feature** appeared to mask the angle-of-attack sensor. Big strike No. 3.

Use of a unqualified
spare part sensor

None of the above should have passed muster. None of the above passed the "OK" pencil of the most junior engineering staff, in fact.

Development in
Low-Cost countries

That's not a big strike. That's a political, social, economic, and technical sin.

Design flaw: missing
redundancy

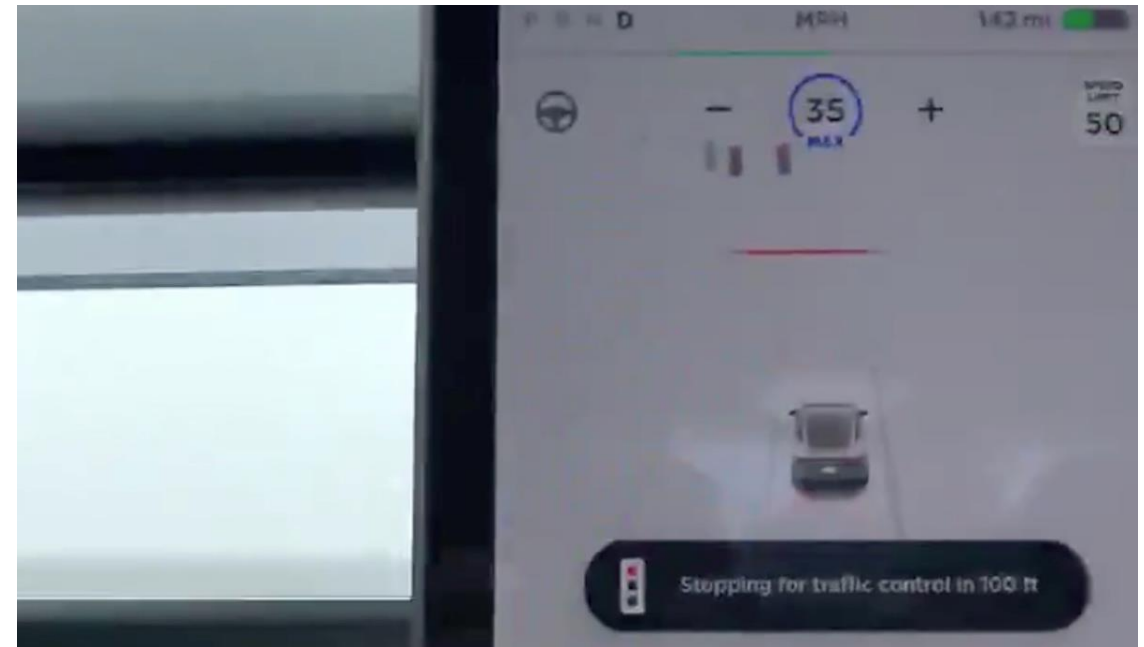
No Training!

<https://spectrum.ieee.org/aerospace/aviation/how-the-boeing-737-max-disaster-looks-to-a-software-developer>

New Feature TESLA auto pilot (27.3.20)

- A new feature of the TESLA Auto Pilot: The car will stop in front of a red traffic light or stop sign even if the driver wants to continue driving.
- The decision will be based on maps and or camera data.
- Good idea or not?

What happens if somebody hacks Google maps? Or jams the GPS signal?



<https://teslamag.de/news/handbuch-tesla-autopilot-rotlicht-stopp-schild-fahrer-befehle-ignoriert-27722>

Typical problems – again and again

Overconfidence
of engineers

Missing system
understanding

No systematic
failure analysis

No independent
QA

Unqualified
software reuse

Low diagnostics

More complex
control
decisions

Ignoring the
“human factor”

Underestimated
races

No proper error
handling

Logical (state)
issues

Key patterns....

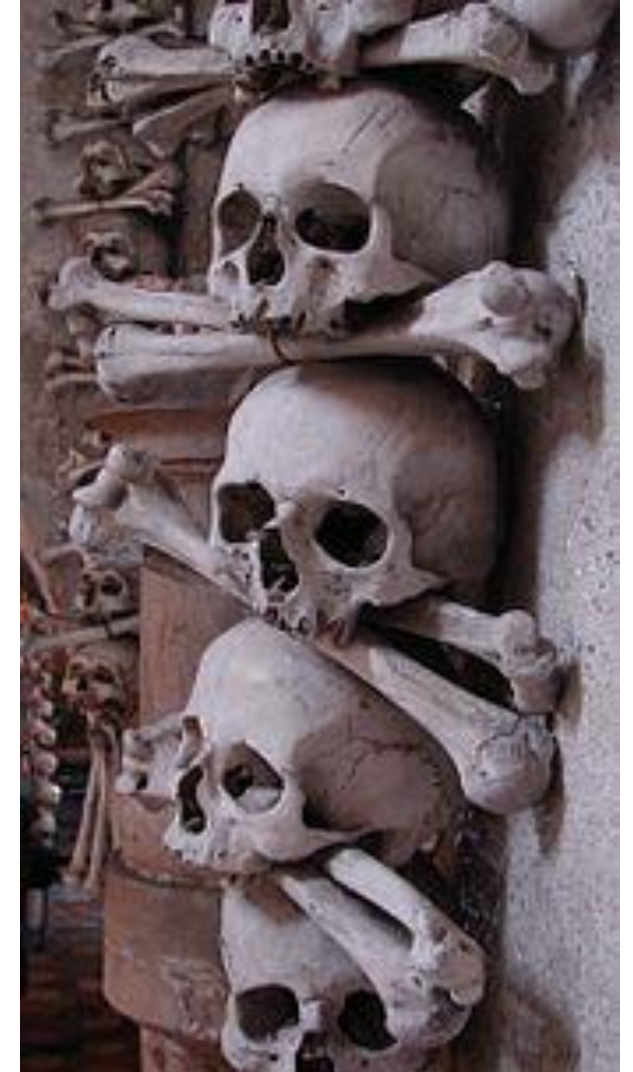
Underestimated hazards and risks!

Believing that it should work does not mean it will work!

Time pressure

Feature pressure

Cost pressure



What means functional safety?

IEC 61508

- **Safety** is the freedom from **unacceptable risk** of physical **injury** or of damage to the **health of people**, either directly, or indirectly as a result of damage to property or to the environment.
- **Functional Safety** is part of the overall safety that depends on a system or equipment operating correctly (i.e. perform a **safety function**) in response to its inputs.

ISO26262

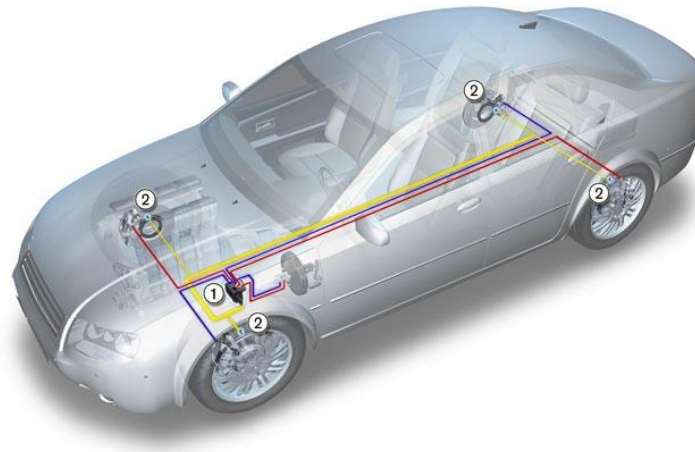
- functional safety: absence of **unreasonable risk** (1.136) due to **hazards** (1.57) caused by **malfunctioning behaviour** (1.73) of **E/E systems** (1.31)

Some examples



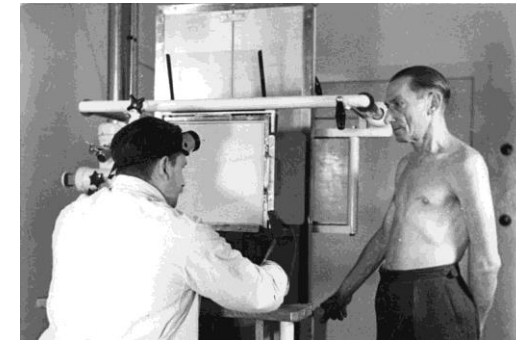
Antiblockiersystem ABS

- 1 Hydroaggregat mit Anbausteuereinrichtung
- 2 Raddrehzahlsensor



Can a malfunction of these systems endanger human life?
Which risks should be considered?
What do these systems have in common, what is different?

Some more examples....



How about such a system?



Safety Requirements
are changing!



Safety function categories

Let's compare the 3 functions Airbag, Electronic Steering, Electronic Brake

	Airbag	Electronic Steering	Electronic Brake
Potential hazard	1) Does not explode during accident 2) Explodes during normal driving	1) Driving does not respond to steering wheel	1) Brake does not react 2) Brakes by itself
Call frequency	Only during accident	Permanently	Regularly
Safe state	Off (Why?)	None	Good question

Protection
function

Aims at reducing the
risk in a dangerous
situation

Safe system
function

Must be safe during
normal operation.

Safe system
function

Must be safe during
normal operation.

Safety function categories – slightly different analytical scope

Protection function

- Must work correctly, when required
- May not interfere if not required
- Typically has a safe state (off)

Safe system function

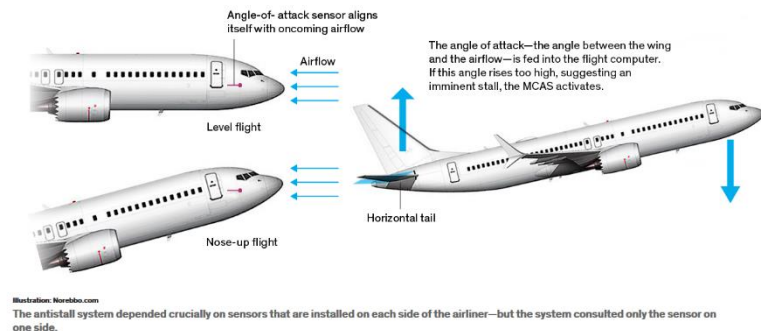
- Must ALWAYS work as specified
- May or may not have a safe state

Often, the second aspect (not interfering when not required) creates the higher risk, as this period is much longer (99%+). Unfortunately, it is easily forgotten!

Reliability and installing a fail safe scenario are major challenges.



How the new Max flight-control system (MCAS) operates to prevent a stall



Which risk is acceptable?

IEC 61508

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- **Functional Safety** is part of the overall safety that depends on a system or equipment operating correctly (i.e. perform a **safety function**) in response to its inputs.

ISO26262

- functional safety: absence of **unreasonable risk** (1.136) due to **hazards** (1.57) caused by **malfunctioning behavior** (1.73) of **E/E systems** (1.31)

Philosophical question

- What is the general risk of life?
- How safe is the system “human”?



- As we are living on a world full of risks, so a Zero-risk system usually does not exist.
- When discussing the term "acceptable risk" the base standard IEC61508 (on which most other safety standards are based) assumes a "natural risk of death" for a 10 year old person of 2×10^{-4} per year.
- Which means, that such a risk value can be considered as tolerable.
- At the same time, a non safety, i.e. QM certified component is considered to have a failure rate (in a low demand mode, i.e. called less than once per year) of 10^{-1} and higher. The range in between is the focus of most safety standards.

Some Definitions

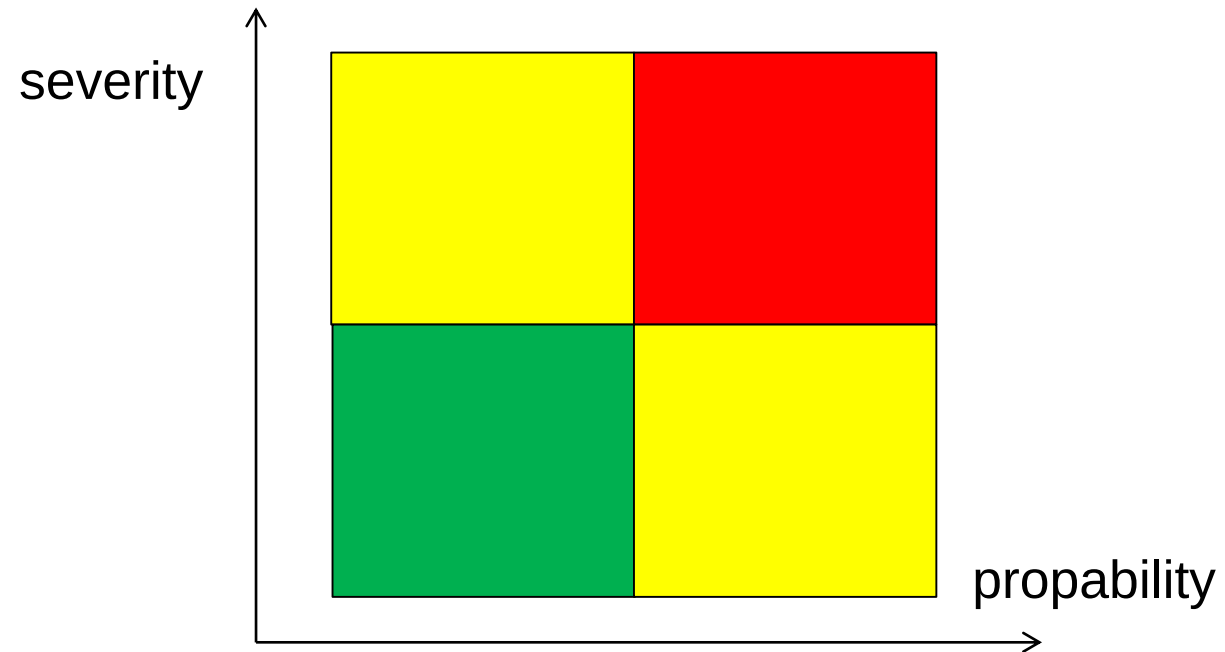
- **Risk:** A risk is a problem that **might happen in the future**
- **Problem:** Inacceptable situation
 - In terms of a safety function: Endangering humans or environment. Also referred to as **hazard**.
- **Hazardous Event:** A hazardous event turns a risk into a problem, e.g. failure of a critical component.

Some more definitions

- **Severity:** Impact of the problem / hazard (economical, environmental,..)
- **Probability:** probability of the hazardous event to occur.
- **Controllability:** probability of the human operator to control the hazardous event, i.e. prevent the hazard to occur.
- **Exposure:** How often is the system exposed to a hazardous event (e.g. ABS only relevant in braking situations)

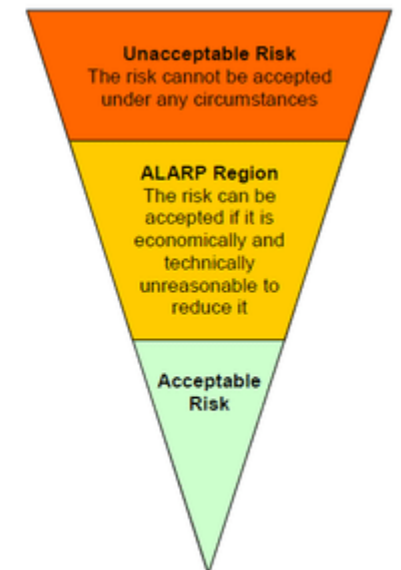
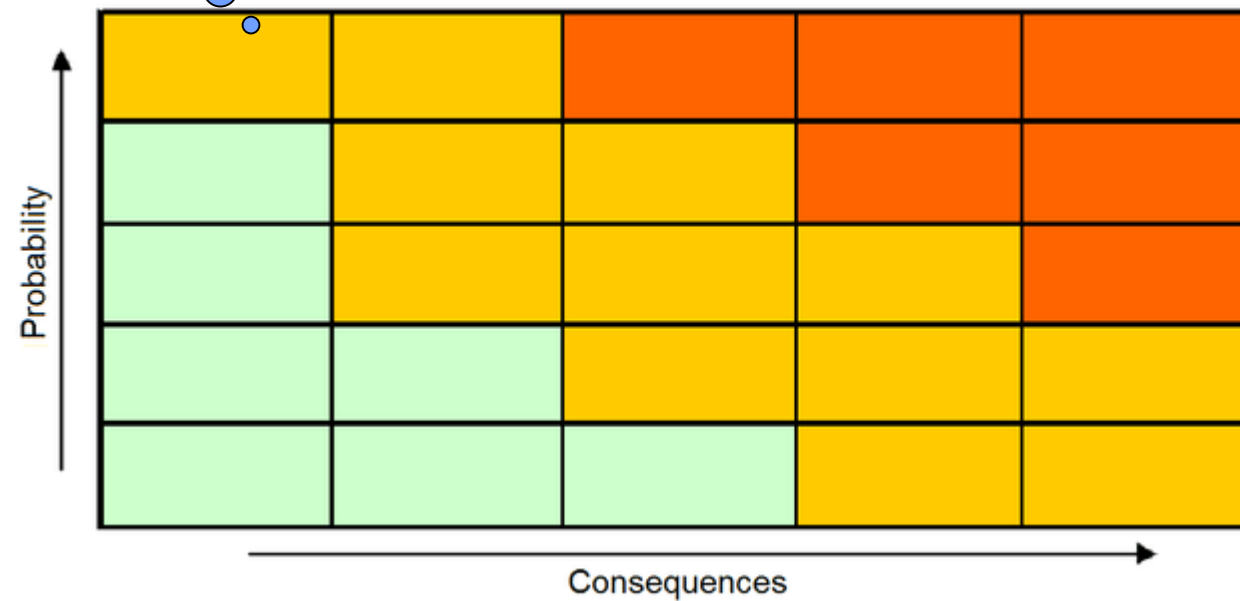
Risk classification - principle

„A risk is a **problem** that **might** happen“



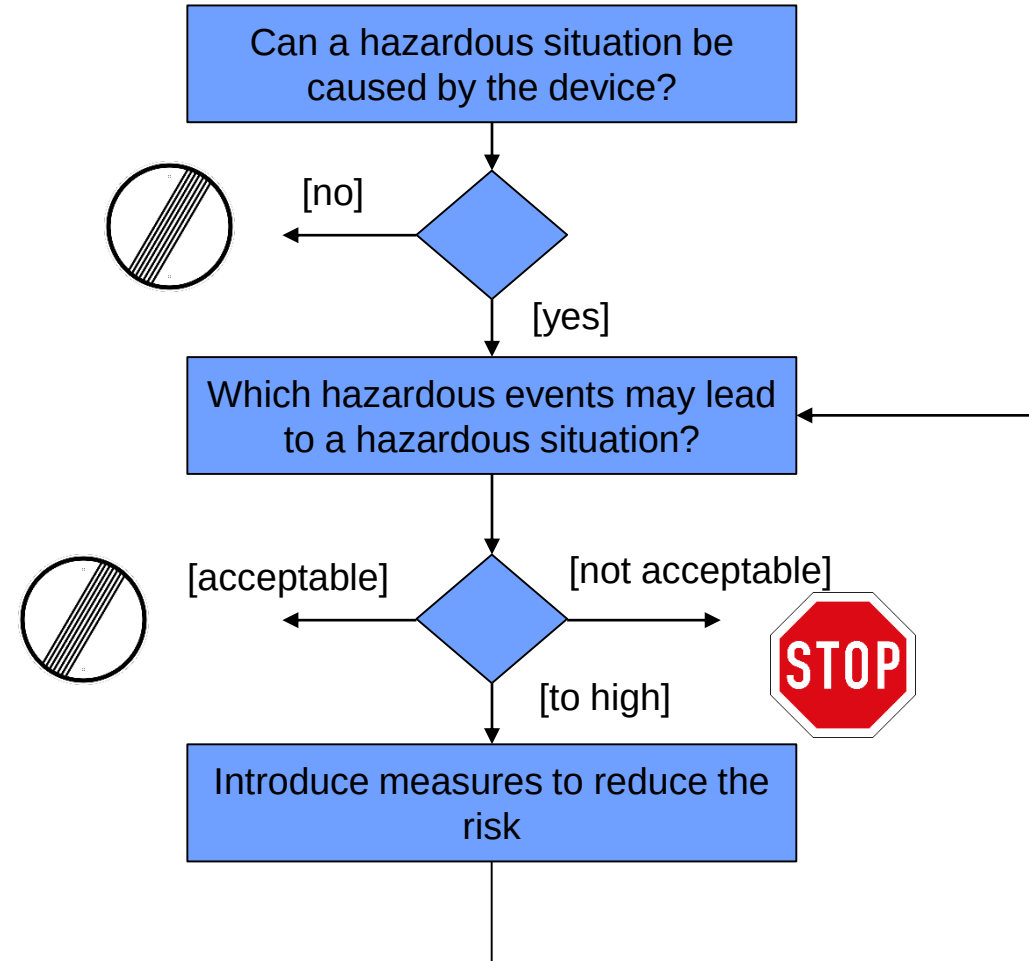
As Low As Reasonable Applicable (ALARP)

This is qualitative. A quantitative concept defining so-called FIT's will be introduced later.



FIT: Failure in time

Hazard and Risk Analysis bicycle brake



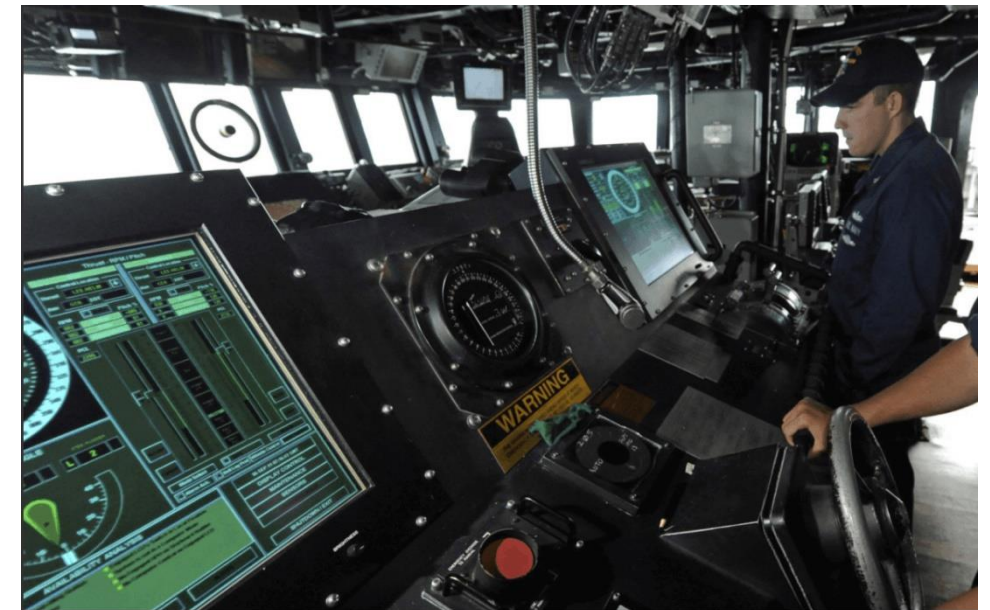
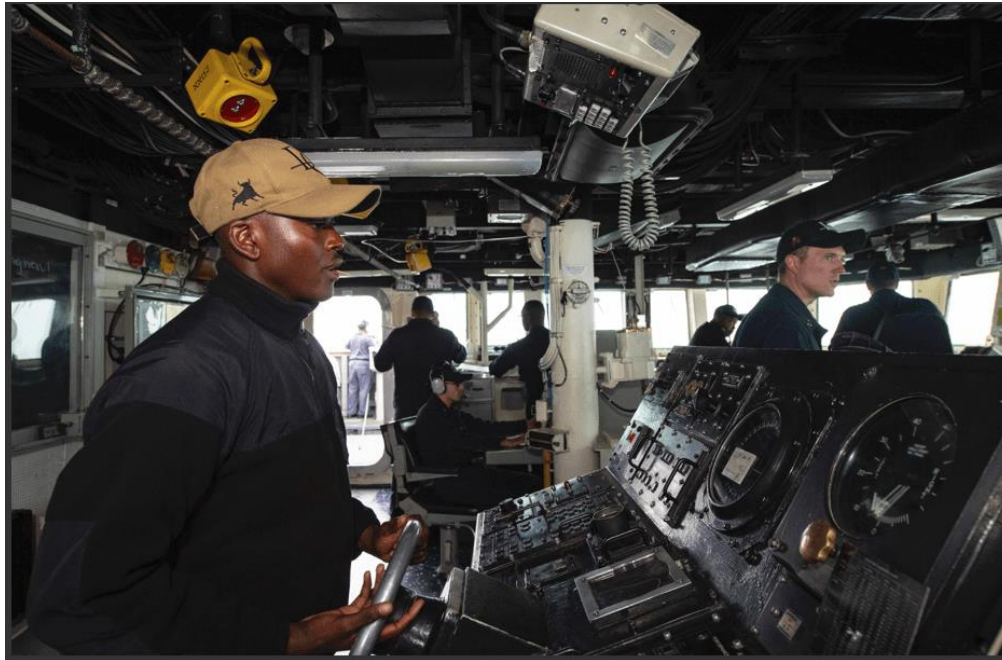
Another Perspective – Cockpit of the Porsche Taycan



Is this a safety case?

Back to the basics

US Navy returns to conventional switches to reduce operational errors



<https://winfuture.de/news,110586.html>

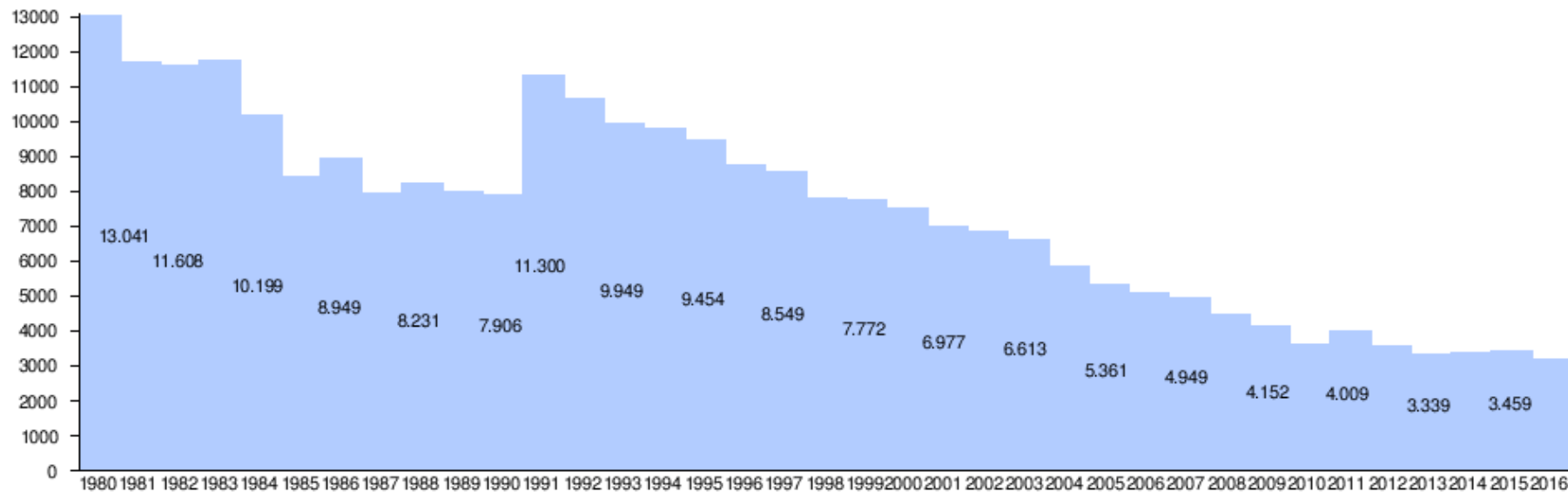
A medical example (infusion pump) – notice something?



Risk evaluation - Do we need autonomous cars?

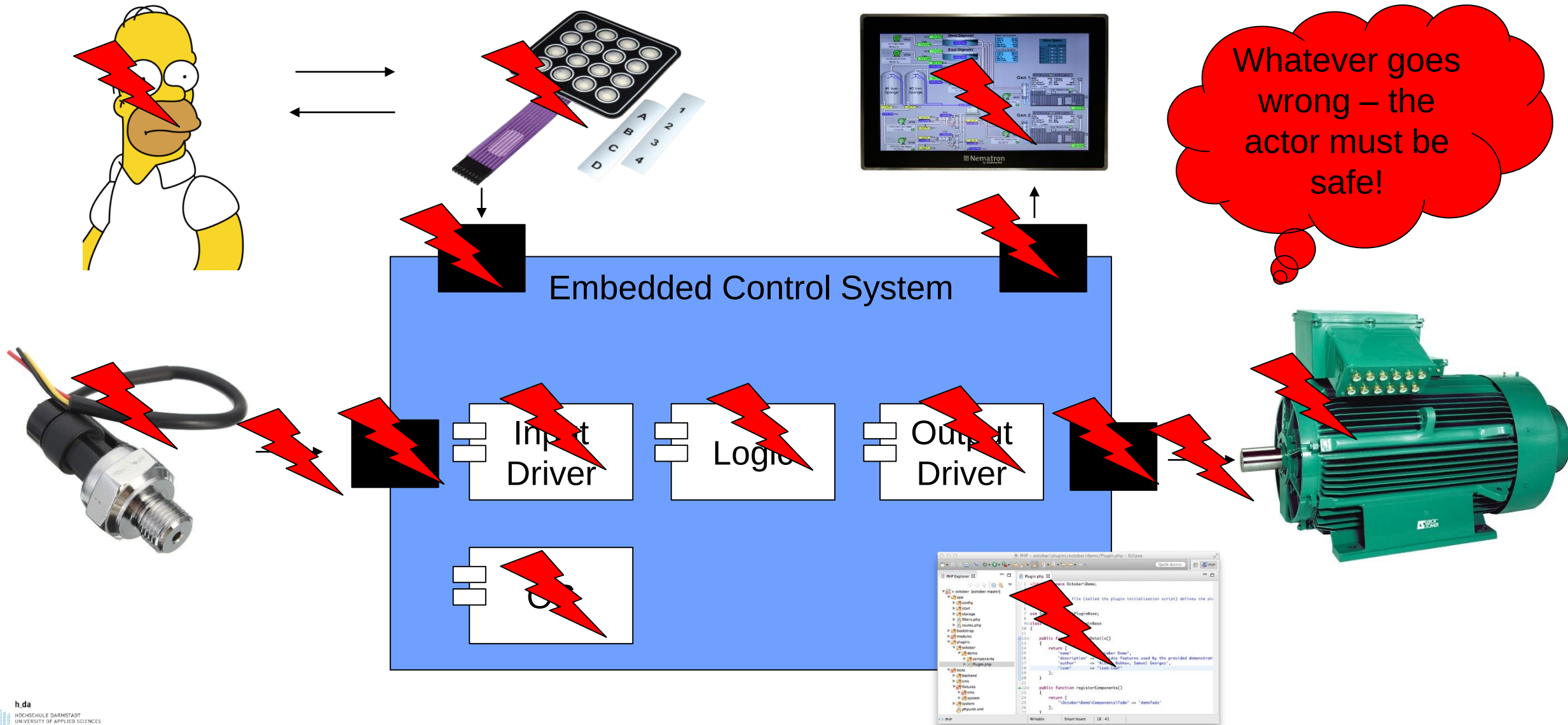
Statement 1: Autonomous cars will reduce the number of deadly accidents by 50%.

Statement 2: Autonomous cars will have to learn to drive. At the beginning, we will very likely see around 1500 deadly accidents caused by autonomous cars.



Source: <https://de.wikipedia.org/wiki/Verkehrstod>

Safety in Embedded Control Systems



Typical Issues (1)

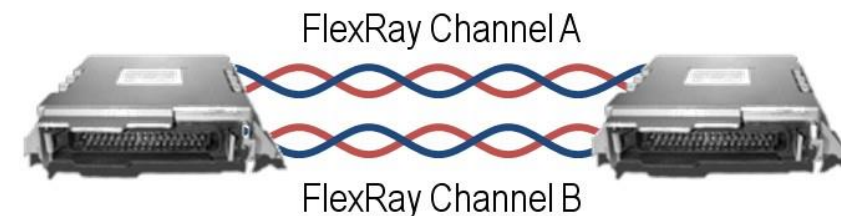
Sensor measures wrong value

- Bad connections
- Execution out of physical limits
- Power supply problems
- Physical wear of
- Wrong sensor configuration
- Environmental disturbances
- Wrong location
- Bad quality
- Wrong measurement principle
- Bug in firmware
- ...



Communication problems

- Wrong protocol
- Environmental disturbances, ESD
- Wrong line setup (e.g. EOL Resistor missing)
- Too long lines, to high baudrate
- CRC not checked
- Wrong wires, bad connectors
- Problem with bus participant (load, collisions, bus heavy)
- ...



Typical Issues (2)

Actor shows wrong behaviour

- Wrong control value
- Wrong load
- Bad connections
- Power supply problems
- Mechanical wear of / bad quality
- Wrong actor configuration
- Environmental disturbances
- Bug in firmware
- ..



Microcontroller hardware

- Wrong pin configuration
- Power supply problems
- Wrong pin load (too high current)
- Wrong external pin circuitry
- ESD issues
- Flipping memory bits
- Sticky pins
- ...



Typical Issues (3)

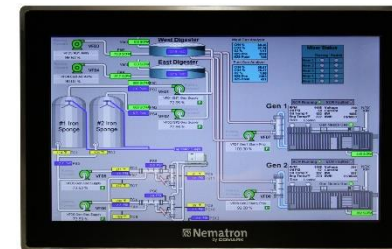
Software

- Misunderstood requirements
- Using invalidated (sensor) data
- Wrong logic
- Wrong operations (data)
- Pointer errors
- Forgotten error handling
- Races, critical regions
- OS Configuration (stack,....)
- Non-qualified build tools
- ...



Human machine interface

- Error prone input device
- No (tactile) feedback (touch)
- Hard to understand information
- Violation of “common concepts” – e.g. red indicates good, green indicates error
- Error messages disappear too quickly
- Non-persistent and secure logging
- ...



So – what do we have to do?

The IEC 61508 „Functional safety for electrical/electronic/programmable safety related systems" defines various methods for:

- Avoid systematic errors during development
- Supervision of the system to detect sporadic faults
- Safe handling of errors and transition into a safe state (if such a state exists)



IEC 61508

Safety = Function + Architecture + Process

Function

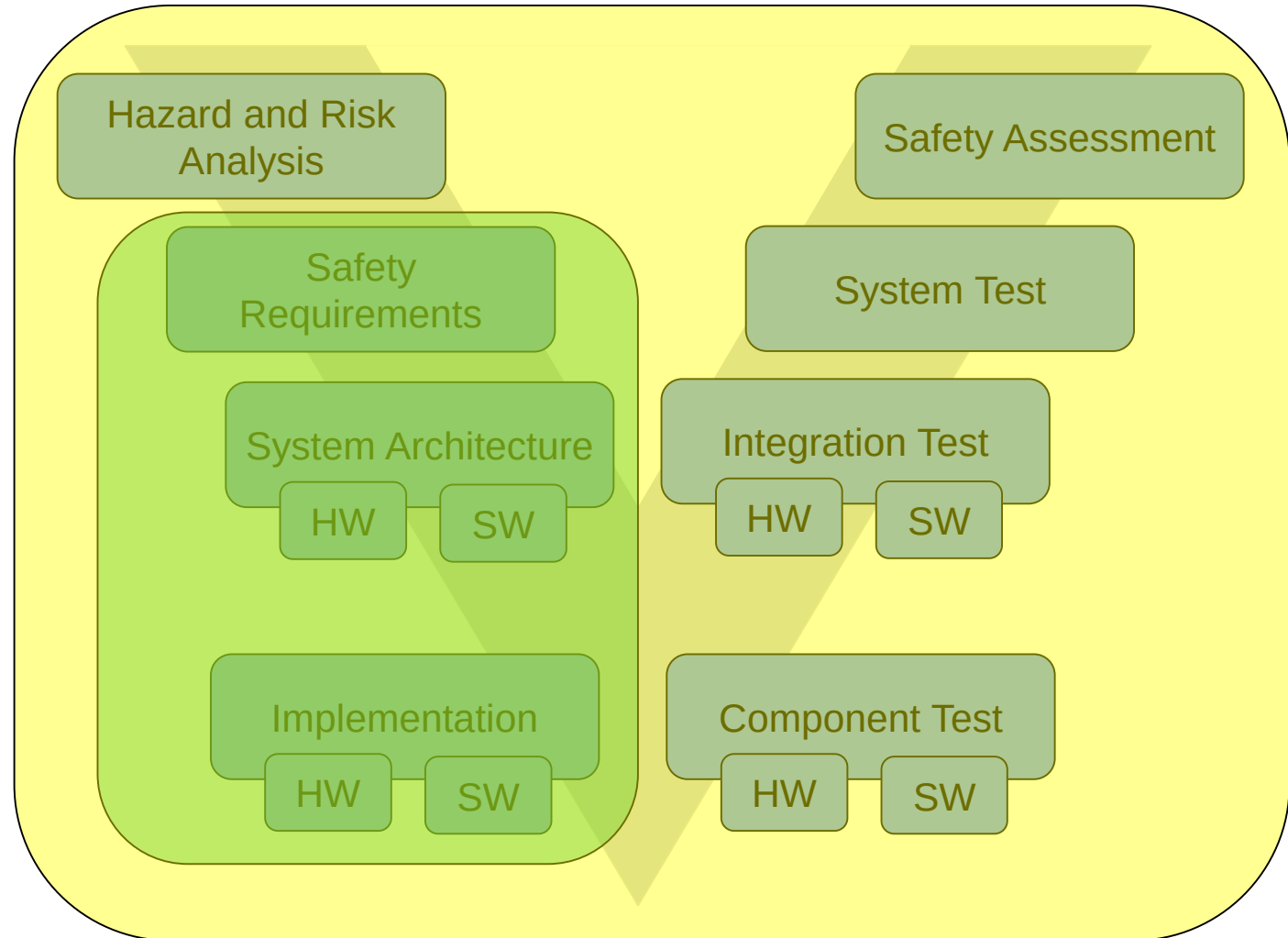
- Risk mitigating protection function
- Supporting the human operator versus taking over control

Architecture

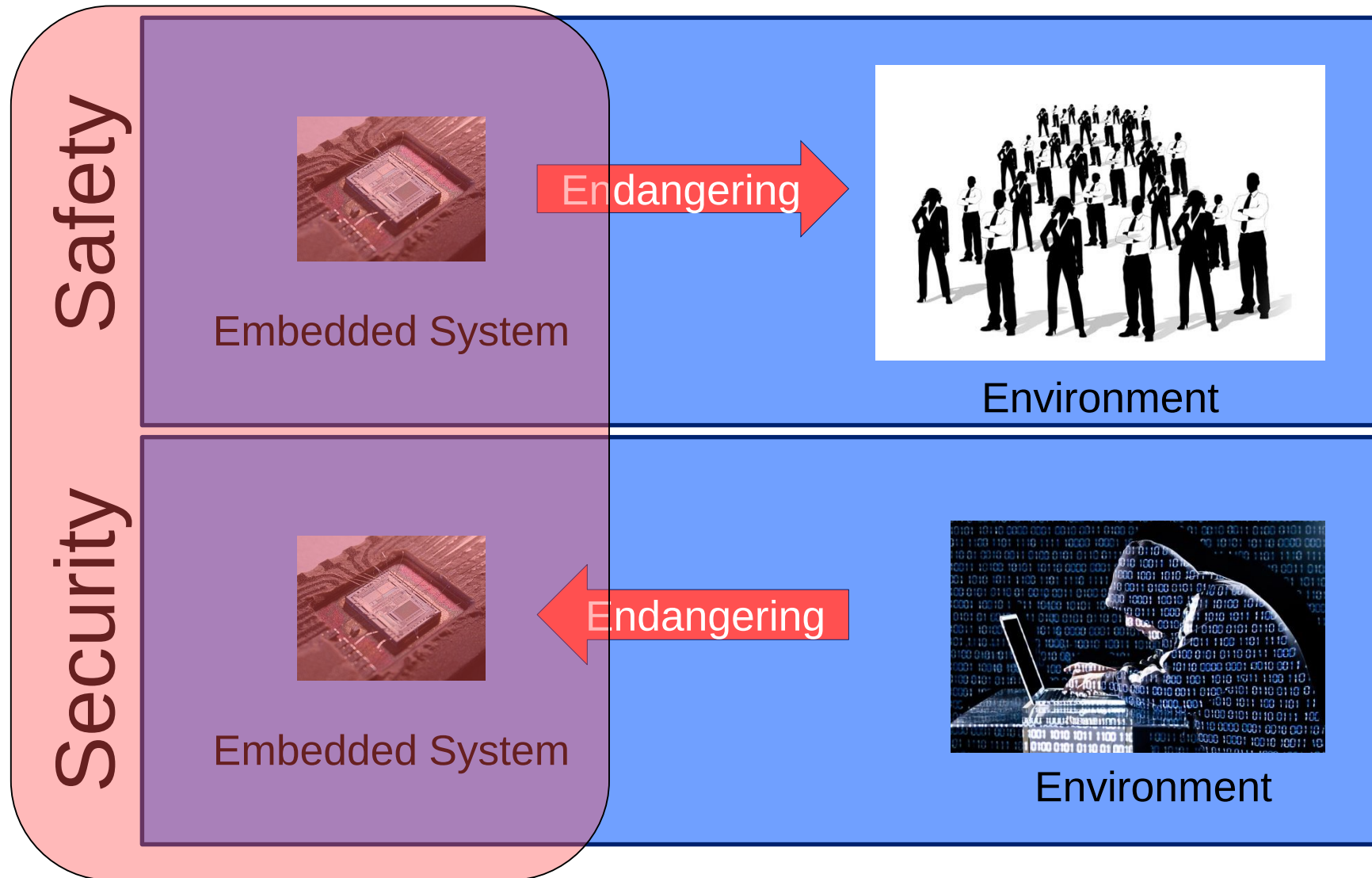
- Choice of correct components and technology
- Error detection and handling
- Redundancy
- ...

Process

- Hazard and Risk Analysis
- Identification and assessment of safety cases and safety requirements
- Architectural modelling
- Use of Design and Coding Pattern
- Test techniques and test coverage
- Safety Assessment
- ...



Safety versus (?) Security



How many bugs are inside a software system having 25.000.000 LOC?

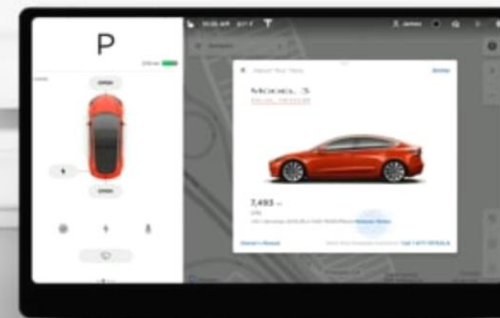
Fast
bugfixes
possible

Software Updates

Our cars regularly receive over-the-air software updates that add new features and enhance existing ones over Wi-Fi.

When updates become available, you'll receive a notification on your center touchscreen display, with the option to install the update immediately or schedule for later. To ensure the fastest and most reliable delivery of software updates, connect your car to Wi-Fi.

Once complete, review the
release notes for details



Software gets
mature at the
customers site

Possible
manipulation by
hackers.

The famous jeep hack (Google jeep hack ;-)



<https://www.youtube.com/watch?v=MK0SrxBC1xs>

Wrapup

- Whenever a technical system endangers humans life, functional safety needs to be considered.
- Control systems become more and more prominent and complex, i.e. the need for safety assessments will grow.
- Safety standards provide process and architectural recommendation to develop safety systems (based on the state of the art when the standard was published).
- Security introduces another dimension when developing open safety systems.